

Math 54 Discussion Worksheet

Name: _____ ID: _____

Problem 1

- a) If $S = \{u_1, \dots, u_m\} \subset \mathbb{R}^n$ is an orthogonal set, then it is linearly independent and thus forms a basis for the space spanned by S .
- b) Let $S = \{u_1, \dots, u_m\}$ be an orthonormal basis for a subspace $W \subset \mathbb{R}^n$. For each $y \in W$, the weights in the linear combination $y = c_1 u_1 + \dots + c_m u_m$ are given by

$$c_i = \frac{u_i \cdot y}{u_i \cdot u_i}.$$

Problem 2

- a) An $m \times n$ matrix has orthogonal columns iff $U^T U = I$.
- b) Prove the following facts about orthogonal matrices: for any $x, y \in \mathbb{R}^n$ we have
- $\|Ux\| = \|x\|$,
 - $(Ux) \cdot (Uy) = x \cdot y$,
 - $(Ux) \cdot (Uy) = 0$ iff $x \cdot y = 0$.

Problem 3

- a) Let U be an $n \times n$ orthogonal matrix. Show that $\det(U) = \pm 1$. Use this to argue that every orthogonal matrix is invertible.
- b) If U and V are orthogonal matrices, then their product UV is also orthogonal.